

Ask, Don't Guess: Learning When to Clarify in Situated Instruction Following

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Abstract

Language agents in situated environments must decide not only what an instruction might mean, but whether it is useful and safe to act on that interpretation. Existing clarification benchmarks often treat this as static ambiguity detection. We instead frame clarification as a situated value-of-information decision under uncertainty, interaction cost, and task consequences. We introduce CLARIFY-TO-ACT, a procedural benchmark in which an agent observes a structured scene and instruction, then either acts immediately or asks one clarifying question before acting. Rewards are deterministic and combine task success, clarification cost, and wrong-action cost. Across different models, direct ask-when-needed prompting improves over guessing but remains miscalibrated: it over-asks when ambiguity is harmless and under-asks when wrong actions are costly. An expected communicative utility (ECU) policy reaches 0.98 utility with no missed or unnecessary clarifications in these main runs. The central lesson is that pragmatic clarification should be evaluated as utility-calibrated interaction, not ambiguity detection alone.

1 Introduction

A situated language agent often receives instructions that are easy to parse but underspecified for action. A household assistant asked to “bring the red mug” may see two red mugs. A file agent asked to “delete the old draft” may see two plausible drafts with very different consequences. A robot asked to “move a spare chair” may see several chairs, any of which would satisfy the request. The agent must decide whether to act now or spend an interaction turn asking what the user means.

This decision is pragmatic: interpretation depends on context, mutual knowledge, informativeness, and consequences (Grice, 1975; Sperber & Wilson, 1986; Clark & Wilkes-Gibbs, 1986; Clark, 1996; Frank & Goodman, 2012; Goodman & Frank, 2016). It is also situated: the same string can require different behavior depending on scene state, affordances, and stakes (Bisk et al., 2020; Min et al., 2024). Ambiguity is sometimes harmless, as when all candidate objects are interchangeable. Context can make an ambiguous request safe to execute. Conversely, a likely interpretation can still require clarification if the wrong action is costly. Clarification therefore resembles a value-of-information problem (Rao & Daumé, 2018; Horvitz, 1999): asking is useful only when the expected benefit of information exceeds the cost of interruption.

We introduce CLARIFY-TO-ACT, a controlled benchmark for this ask-or-act decision. Each episode contains a structured scene, an instruction, candidate user intents with priors, a hidden user intent, and task costs. The agent may act or ask. If it asks, a simulated user answers from the hidden intent and the agent then acts. Because the environment computes rewards automatically, CLARIFY-TO-ACT supports cheap and reproducible experiments with hosted models and transparent controllers.

Contributions. We introduce a 1,400-episode benchmark for situated clarification; formalize clarification as expected communicative utility (ECU); compare prompting, private-reasoning prompting, transparent controllers, and an expected-utility wrapper. The key

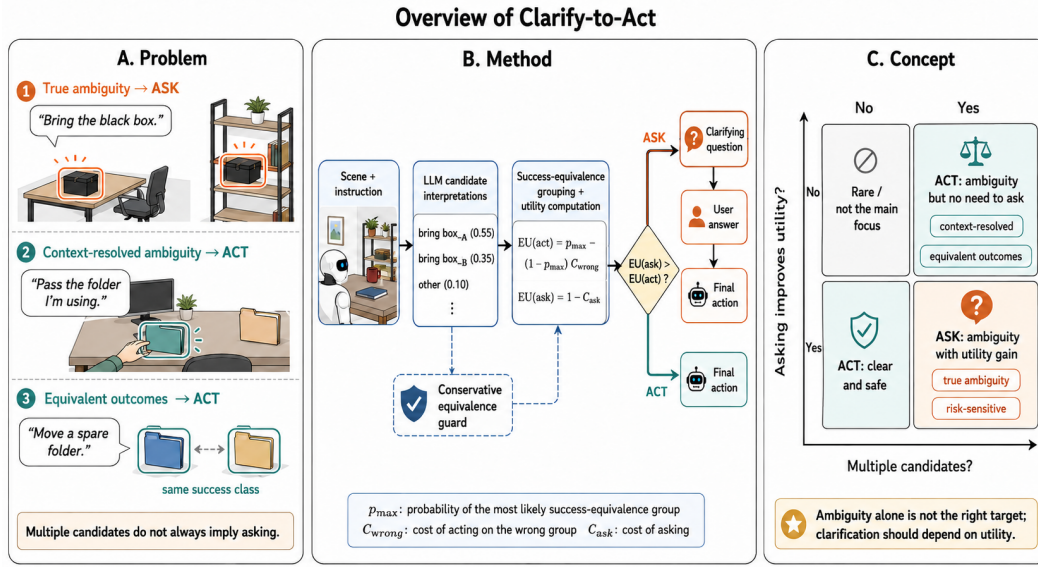


Figure 1: Overview of CLARIFY-TO-ACT. The benchmark separates ambiguity that requires asking from ambiguity resolved by context or outcome equivalence, and the ECUmethod asks only when clarification improves expected task utility.

43 result is that explicit utility calibration gives a stable rule for when communication is worth
44 its cost.

45 2 Clarify-to-Act

46 Each CLARIFY-TO-ACT episode is generated from a compact scene grammar and contains a
47 hidden intent sampled from candidate interpretations. The agent observes the scene and
48 instruction, but not the hidden intent. It outputs either

$\{\text{type: ACT, action: } a, \text{target_id: } o\}$

49 or

$\{\text{type: ASK, question: } q\}.$

50 If it asks, the simulated user returns a state, location, or ownership answer associated with
51 the hidden target, and the agent then chooses a final action. Episodes are capped at one
52 question to focus evaluation on the first-turn ask-or-act decision.

53 The canonical dataset contains 1,400 episodes. Splits are balanced across the five categories
54 in Table 1; the overall oracle ask rate is 0.50.

55 **Reward and oracle.** The reward is

$$R = \mathbb{1}[\text{success}] - C_{ask}\mathbb{1}[\text{asked}] - C_{wrong}\mathbb{1}[\text{wrong}].$$

56 Success is defined by matching the hidden intent’s success-equivalence class, not necessarily
57 by matching a target identity. Let $p_{\max}$ be the prior mass of the best success-equivalence
58 class. Acting has expected utility $EU(\text{act}) = p_{\max} - (1 - p_{\max})C_{wrong}$, while asking has
59 $EU(\text{ask}) = 1 - C_{ask}$. The oracle asks when $EU(\text{ask}) > EU(\text{act}) + \epsilon$. We report net utility,
60 success, ask rate, missed clarification, and unnecessary clarification.

61 **Simulated user and leakage controls.** The simulated user is deterministic, which makes
62 replay exact. Hidden owner names are redacted from prompts, hidden-owner objects
63 receive neutral state labels, and target identifiers are neutral. A simulated-user audit verifies
64 that deterministic answers resolve the hidden success class in all generated oracle-ask cases
65 and all actual asked rows.

Category	Example instruction	Oracle ask	What it tests
Referential	Bring the black box	1.00	Matching objects; wrong target matters
Context-resolved	Pass the folder I’m using	0.00	Scene context resolves the target
Equivalent outcome	Move a spare folder	0.00	Targets share one success class
Risk-sensitive	Delete the old draft	1.00	Wrong action is costly despite high prior
Preference/social	Put my cup away	0.50	Ownership visible vs. hidden

Table 1: Benchmark categories. Every category contains multiple candidate interpretations; the oracle ask rate follows expected communicative utility rather than surface ambiguity.

Model / split	Plain Ask	CoT Ask	ECU	Main prompt failure
GPT-5-mini	0.63	0.63	0.98	under-asks risk/reference cases
GPT-5.4-mini	0.87	0.86	0.98	over-asks resolved cases
GPT-5.5	0.85	0.94	0.96	plain prompt under-asks high-risk cases

Table 2: Stronger models and private reasoning help, but explicit expected-utility calibration is the most consistent strategy.

3 Methods and Experiments

We compare four families of policies. DIRECTACT never asks. ASKALWAYS and RAWAMBIGUITY ask whenever ambiguity is present, measuring over-clarification. ASK-NEEDED prompts the model to ask only when acting could plausibly fail; a private-reasoning variant encourages hidden deliberation before the final JSON answer. The ECU policy first asks the model to propose candidate interpretations, probabilities, context-resolution flags, equivalence flags, and risk estimates. Deterministic code then groups candidates into success-equivalence classes and computes $EU(ask) - EU(act)$. If the margin exceeds 0.1, the model generates a clarifying question; otherwise the agent acts. Model-declared equivalence is accepted only when supported by explicit cues such as “spare,” “any,” or “one of.”

We also test paraphrased instructions, terser answer styles, reversed object order, compact natural-language scene descriptions, held-out ambiguity mixes, component ablations, candidate-probability diagnostics, and cost re-scoring.

4 Results

We emphasize three takeaways.

4.1 Takeaway 1: utility calibration improves ask-or-act behavior

Table 2 summarizes the main results. On GPT-5-mini, direct ASK-NEEDED prompting improves over never asking but misses more than half of the oracle clarification cases. ECU reaches 0.98 utility with full task success and no ask errors. Stronger models narrow the gap. GPT-5.4-mini prompts over-clarify many oracle-act cases, while GPT-5.5 plain prompting under-clarifies risk-sensitive cases. ECU keeps the ask rate aligned with the oracle and has zero missed and zero unnecessary clarifications in the main current-model runs.

4.2 Takeaway 2: ambiguity and uncertainty are not enough

All canonical test episodes have multiple candidate interpretations, yet half are oracle-act cases and half are oracle-ask cases. A raw ambiguity detector asks on everything and pays

Matched situation	Oracle	Why
Two bring targets; one active/reachable	ACT	Context makes one candidate utility-dominant
Two visually matching bring targets	ASK	Candidates have different success classes
“Put my mug away” with visible ownership	ACT	User identity and owner resolve “my”
“Put my mug away” with hidden ownership	ASK	The scene lacks ownership information
High-entropy spare-object choice	ACT	All candidates share one success class
High-prior delete-file choice	ASK	Wrong-action cost makes clarification valuable

Table 3: Why surface ambiguity is not the right target. Similar-looking ambiguity can require opposite actions once context, equivalence, and risk are included.

Robustness check	Main finding
Paraphrase/style stress	ECU remains near 0.98 utility; plain ASK-NEEDED still misses many oracle-ask cases
Reversed object order	ECU’s ask/act decisions are unchanged on the shared GPT-5.4-mini subset
Natural-language scene format	ECU keeps full task success with only one unnecessary clarification
Held-out ambiguity mix	Expected-utility reasoning transfers better than the tuned controller under category shift
Cost re-scoring	ECU keeps a positive paired utility advantage across tested costs

Table 4: Robustness summary. The detailed numbers are less important than the pattern: utility calibration is stable under bounded prompt, scene, category, and cost perturbations.

unnecessary clarification cost whenever context or outcome equivalence already resolves the instruction. An uncertainty-only controller improves over raw ambiguity but still misses value-relevant structure: equivalent outcomes make entropy harmless, while high wrong-action cost can make clarification valuable despite a high top prior.

4.3 Takeaway 3: ECU separates interpretation from decision-making

ECU works by separating semantic interpretation from the ask decision. The LLM proposes plausible interpretations and rough probabilities; deterministic code computes whether asking is worth the cost. Ablations support this mechanism. If every model-declared equivalence flag is trusted, utility drops because the model sometimes collapses distinct referents. If equivalence is never collapsed, success remains high but the agent asks many unnecessary questions. The conservative equivalence guard is therefore essential.

The robustness checks in Table 4 reduce several simple alternative explanations. The result is not driven by object order, by JSON syntax alone, or by one narrow reward setting. Candidate-probability diagnostics also support a bounded mechanism claim: ECU does not require perfect hidden-intent prediction; it needs candidate sets and rough relative probabilities good enough for the utility margin to track the oracle margin.

5 Qualitative Analysis

The most informative failures are first-turn calibration failures. Prompted ASK-NEEDED often over-clarifies equivalent-outcome instructions: when asked to move a spare

folder to the table, it asks which folder even though any spare folder succeeds. It also under-clarifies high-risk actions: in delete-file cases, it sometimes acts immediately despite high wrong-action cost. Preference/social cases expose a related failure: the model may treat “my” as resolved even when ownership is hidden. Context-resolved cases expose the opposite problem: the model may ask even when active workspace context identifies the intended object. ECU avoids these failures because it explicitly asks whether information changes expected utility.

6 Related Work

Clarification and mixed initiative. CLAM, AmbigQA, and CLAMBER study selective clarification and ambiguous information needs (Kuhn et al., 2022; Min et al., 2020; Zhang et al., 2024). Conversational search and value-of-information work ask clarifying questions to resolve underspecified needs (Aliannejadi et al., 2019; Rao & Daumé, 2018), while mixed-initiative interfaces emphasize interrupting only when intervention is valuable (Horvitz, 1999). Human-robot dialogue work shows that clarification can improve grounded language understanding (Thomason et al., 2019), and recent work compares human and LLM behavior under referential ambiguity (Madge et al., 2025).

Situated and pragmatic language. Situated language understanding links utterances to perception, affordances, and actions (Bisk et al., 2020; Min et al., 2024). Embodied benchmarks evaluate navigation, household tasks, affordance grounding, multi-modal dialogue, and collaborative drawing (Tellex et al., 2011; Shridhar et al., 2020; Padmakumar et al., 2022; Ichter et al., 2023; de Vries et al., 2017; Kim et al., 2019). Classic pragmatics and Rational Speech Act models treat meaning as context-sensitive inference over speakers, listeners, and utilities (Grice, 1975; Sperber & Wilson, 1986; Clark & Wilkes-Gibbs, 1986; Clark, 1996; Frank & Goodman, 2012; Goodman & Frank, 2016; Monroe et al., 2017; Golland et al., 2010). Emergent-communication and social-agent work similarly study communication as goal-directed interaction (Lazaridou et al., 2017; Zhou et al., 2024). CLARIFY-TO-ACT is narrower but more controlled: the environment computes success and clarification utility directly.

7 Limitations

CLARIFY-TO-ACT is synthetic and text/JSON-based. It isolates ambiguity, context, equivalence, ask cost, and wrong-action cost, but does not test perception, long-horizon planning, multimodal grounding, or real human interaction. The simulated user answers deterministically; audits verify that these answers resolve the hidden success class, but this is not a human-response study. Finally, real systems would need to estimate priors and costs from user preferences, domain risk, and environment state.

8 Conclusion

Clarification is not merely a response to ambiguity. It is a situated decision about whether information is worth the cost of asking for it. CLARIFY-TO-ACT provides a controlled benchmark for this decision. Across offline diagnostics and evaluations, expected communicative utility improves ask-or-act behavior by asking when clarification changes the expected outcome and acting when it does not. Pragmatic clarification should therefore be evaluated as utility-calibrated interaction policy rather than static ambiguity detection alone.

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